



COMPUTER ORGANIZATION

Topic: Microoperations For NIMCET [Date: 27.04.2026]



01. If the control function is given as:
 $P: R2 \leftarrow R1, Q: R3 \leftarrow R2$
What happens if both P and Q are enabled in the same clock cycle?
A. R3 gets old R2
B. R3 gets new R2
C. Undefined behavior
D. No transfer
02. Which of the following ensures no bus conflict during register transfer?
A. Multiple sources enabled
B. Tri-state buffer control
C. Multiple destinations
D. Parallel ALU
03. RTL statement: $R1 \leftarrow R1 + R2$ implies:
A. Only transfer
B. Only logic
C. Arithmetic + transfer
D. Memory operation
Operation happens in ALU, result stored $\rightarrow$ both arithmetic and transfer.
04. Conditional transfer $P: R1 \leftarrow R2$ depends on:
A. Bus width
B. Control signal P
C. Register size
D. Memory
05. Which is true for RTL?
A. Sequential execution
B. Parallel execution possible
C. Only arithmetic
D. Only logic
06. If multiple registers attempt to write to the same bus:
A. Valid transfer
B. Faster transfer
C. Bus contention
D. No effect
07. Microoperations occur at:
A. Instruction level
B. Clock pulse level
C. OS level
D. Compiler level
08. Which is NOT part of register transfer?
A. Source
B. Destination
C. Control
D. Cache mapping
09. In RTL, $\leftarrow$ indicates:
A. Equality
B. Assignment
C. Comparison
D. Logical op
10. Bus width determines:
A. Speed only
B. Data size transferred
C. Memory size
D. ALU size
11. Parallel transfer requires:
A. Single wire
B. Multiple lines
C. Memory
D. Cache
12. Register transfer without clock:
A. Possible
B. Impossible
C. Depends
D. Undefined
13. In 2's complement subtraction, $A - B$ is done as:
A. $A + B$
B. $A + B' + 1$
C. $A - B'$
D. $A + B'$
14. Overflow in signed addition occurs when:
A. Carry out = 1
B. MSB carry mismatch
C. Result zero
D. No carry
Overflow = carry into MSB $\neq$ carry out.
15. Increment operation can be implemented using:
A. Full adder
B. Half adder
C. Comparator
D. Decoder
16. If $A = 0111$ and $B = 0001$ (4-bit), $A + B$ results in:
A. 1000
B. Overflow
C. Both
D. None
17. Which operation does NOT affect carry flag?
A. Addition
B. Subtraction
C. Increment
D. Logical AND
18. Arithmetic shift left is equivalent to:
A. Divide by 2
B. Multiply by 2
C. No change
D. Rotate
19. In subtraction using 2's complement, borrow is handled by:
A. Ignoring carry
B. Using carry
C. XOR
D. OR

20. Which flag indicates zero result?
A. Carry B. Zero
C. Sign D. Overflow
21. Adding two negative numbers may cause:
A. Carry only B. Overflow
C. No change D. Logical error
22. ALU performs operations on:
A. Memory B. Registers
C. Cache D. Disk
23. Decrement can be done by:
A. Add -1 B. Subtract 1
C. Both D. None
24. Arithmetic ops are:
A. Bitwise B. Numeric
C. Logical D. Transfer
25. Which is fastest?
A. Memory op B. Register op
C. Disk op D. Cache miss
26. Which operation is reversible?
A. AND B. OR
C. XOR D. NOT
27. Masking is done using:
A. OR B. AND
C. XOR D. NOT
28. Clearing a bit uses:
A. AND with 0 B. OR with 1
C. XOR D. NOT
29. Setting a bit uses:
A. AND B. OR
C. XOR D. NOT
30. Toggle operation:
A. AND B. OR
C. XOR D. NOT
31. XOR of same bits gives:
A. 1 B. 0
C. Undefined D. Carry
32. Which operation produces complement?
A. AND B. OR
C. NOT D. XOR
33. Logical ops affect:
A. Carry B. Overflow
C. Bits D. Memory
34. Bitwise AND of 1010 & 1100:
A. 1000 B. 1110
C. 0010 D. 1010
35. OR of 1001 | 0101:
A. 1101 B. 0000
C. 1111 D. 1010
36. XOR of 1111 ^ 1111:
A. 1111 B. 0000
C. 1010 D. 0101
37. Logic ops are done in:
A. CU B. ALU
C. Bus D. RAM
38. Which is NOT logic op?
A. AND B. OR
C. ADD D. XOR
39. Logical shift right inserts:
A. 1 B. 0
C. Sign D. Carry
40. Arithmetic right shift preserves:
A. LSB B. MSB
C. Carry D. Zero
41. Circular shift left of 1001:
A. 0011 B. 0010
C. 0001 D. 0100
42. Left shift of 0101:
A. 1010 B. 0010
C. 1100 D. 1000
43. Right shift of 1000:
A. 0100 B. 0001
C. 0010 D. 0000
44. Arithmetic left shift may cause:
A. Underflow B. Overflow
C. No change D. Rotate
45. Shift operations are controlled by:
A. ALU B. CU
C. Bus D. Memory
46. Rotation differs from shift because:
A. No loss of bits B. Faster
C. Uses ALU D. None
47. Logical shift is used for:
A. Signed numbers B. Unsigned numbers
C. Floating D. Memory
48. Arithmetic shift is used for:
A. Unsigned B. Signed
C. Floating D. Cache
49. Shift register performs:
A. Storage B. Shift
C. Transfer D. All
50. Right shift equals division by:
A. 2 B. 4
C. 8 D. 16